

Chapter 9

Tangency, Power, Radical Axis

In this chapter, that marks the end of the theoretical material as it pertains to the inversion of the real plane with respect to a circle, we tie the loose ends by looking at a few notions that will come in handy in the future.

First, we investigate how inversion acts on the points of tangency between a straight line and a circle and between two circles. Then, we take a look at how inversion acts on the so-called orthogonal circles, the objects that we have already met. Then, we introduce the notion of the power of a point with respect to a circle and, lastly, we will learn what the radical axis of two circles is.

Points Of Tangency

Theorem 37. If a straight line, l , and a circle, q , touch each other at a point, T , distinct from the center of inversion, O , then their images, l' and q' , under an inversion of the ambient plane with respect to a circle, $c(O, r)$, will also touch at a point, T' , that is the image of the point of tangency, T , of the preimage

straight line, l , and the preimage circle, q , under the same inversion with respect to the said circle, $c(O, r)$ and with the same power.

Theorem 38. If two circles, q_1 and q_2 , touch each other at a point, T , distinct from the center of inversion, O , then their images, q'_1 and q'_2 , under an inversion of the ambient plane with respect to a circle, $c(O, r)$, will also touch at a point, T' , that is the image of the point of tangency, T , of the said preimage circles, q_1 and q_2 , under the same inversion with respect to the said circle, $c(O, r)$ and with the same power.

Proofs. The proofs of the above two theorems follow from the combination of the following two facts applied to two interacting curves:

- the (magnitudes of the) angles between the said curves under an inversion with respect to a circle are preserved
- the point of tangency of the said curves, by definition, is unique

From the preservation of the magnitudes of the angles between the curves under an inversion with respect to a circle property, see the Theorems 32 through 36, it follows that if two given original curves are positioned with respect to each other under a certain angle, whatever that angle actually is, then the inverse images of these original curves will also be positioned with respect to each other under the same, by magnitude, angle.

When two curves touch each other then the said angle between them happens to be zero radians but under an inversion with respect to a circle such an angle will be preserved nonetheless.

The Theorem 6 tells us that the inverse image of an original point, whatever that point actually is, is unique. Therefore, if an original point of tangency, T , is unique, which, by definition, it certainly is, then its image, T' , under an inversion of the parent plane with respect to a circle will be unique also.

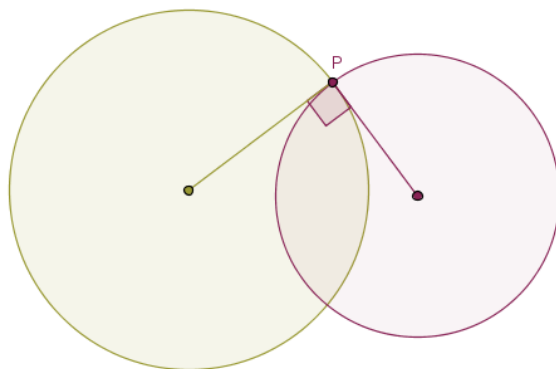
Hence, if T is a point that simultaneously belongs to two touching original curves then its inverse image T' will also simultaneously belong to two touching inverse images of the said original curves. $\square$

Note that the above two theorems apply to an inversion of a plane with both positive and negative powers, with the understanding, of course, that the power of inversion at hand, whatever it is, is fixed.

Orthogonal Circles

Definition 5. From the Definition 4 it follows that any two circles are **orthogonal** at a point, P , or any two circles **intersect orthogonally** at a point, P , if and only if the angle between the tangent straight lines drawn to both circles through the common point, P , is right.

In practical terms, then, any two circles are orthogonal if and only if the angle between their radii at any intersection point is right (Figure 9.1):



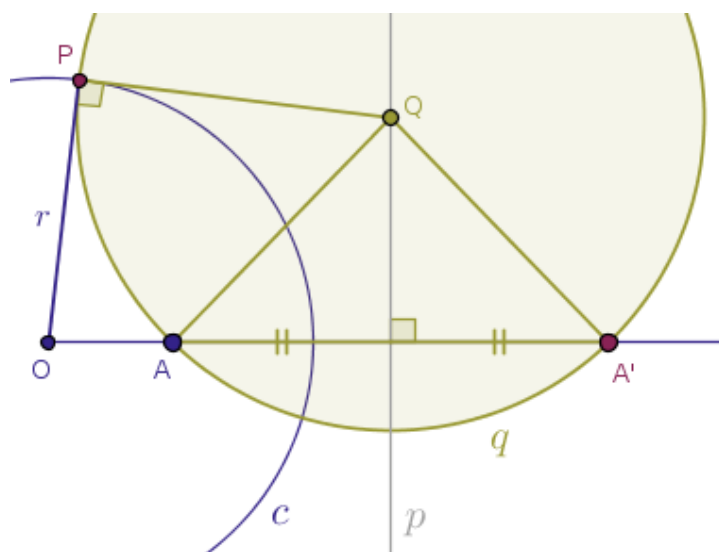
From [Euclid's "Elements" Book 3 Proposition 36](#) and [Book 3 Proposition 37](#) two useful properties of orthogonal circles follow right away.

Theorem 39. Any circle, q , that passes through any two points, A and A' , that are the inversive images of each other with respect to a circle, $c(O, r)$, with positive power is orthogonal to that circle of inversion, c .

Proof. From the Condition 3 of Definition 1 for the points A and A' we have:

$$|OA| \cdot |OA'| = r^2$$

The locus of the centers of all the circles that lie in the plane of the circle of inversion, c , and pass through the points A and A' (Figure 9.2):



must, by definition, be the straight line, p , that is perpendicular to the line segment AA' and that cuts that line segment in half.

Pick an arbitrary point, Q , on that perpendicular bisector, p . Construct a circle, q , centered at that arbitrary point, Q , with the radius whose length is equal to the distance from the chosen point to the given image points, $|QA| = |QA'|$. Let P be one point where the circles c and q intersect.

Since, on the one hand, the point P is located on the circumference of the circle c , it is the case that $|OP| = r$ and, hence:

$$|OA| \cdot |OA'| = r^2 = |OP|^2$$

But since, on the other hand, the point P is also located on the circumference of the circle q , the earlier mentions Euclid's "*Elements*" Book 3 Proposition 37 is applicable to the points O , A and A' with respect to the circle q , where the straight line through the points O and A' plays the role of a secant of the circle q .

Hence, according to that proposition, because the last relation above holds, the straight line OP is perpendicular to the straight line PAQ , from where the conclusion of the theorem follows. $\square$

Theorem 40 (the converse of the Theorem 39). If any two circles, $c(O, r)$ and $q(Q, R)$, are orthogonal then the two points of intersection of any secant of the circle q through the point O are the inverses of each other with respect to the circle c with positive power and vice versa. The two points of intersection of any secant of the circle c through the point Q are the inverses of each other with respect to the circle q with positive power.

Proof. Reusing the Figure 9.2 from the previous theorem, here we are given by the premise that the straight line OP is tangent to the circle q and that the point O is located outside of the circle q .

Hence, for any secant straight line of the circle q that passes through the point O , by the earlier mentioned Book 3 Proposition 36, we will have:

$$|OP|^2 = |OA| \cdot |OA'|$$

But it is the case that $|OP| = r$. Hence:

$$r^2 = |OA| \cdot |OA'|$$

and, since the points O , A and A' are collinear and the points A and A' are located on the same side of the point O , it follows that the points A and A' are indeed the images of each other under the inversion of the parent plane with respect to the circle c with positive power.

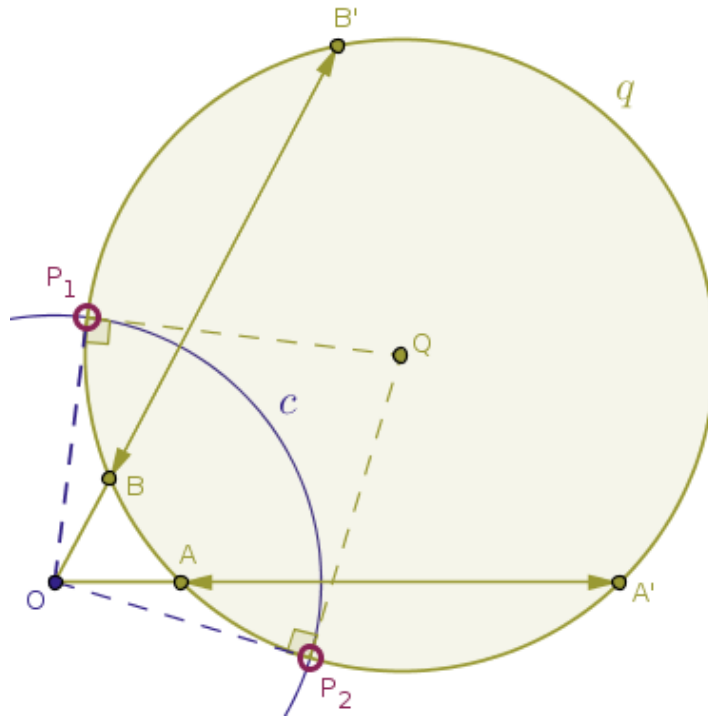
Applying all of the above deductions almost verbatim to an arbitrary secant straight line of the circle c passing through the point Q , we prove that if the circles c and q are orthogonal then the two points of intersection of any secant of the circle c through the point Q are inverses of each other with respect to the circle q with positive power. $\square$

From the above two theorems the next one follows right away.

Theorem 41. If two circles, $c(O, r)$ and $q(Q, R)$, are orthogonal then the circle c is invariant under the inversion of the parent plane with positive power with respect to the circle q and the circle q is invariant under the inversion of the same plane with positive power with respect to the circle c .

Proof. Let us consider the inversion of the circle q with respect to the circle c with positive power.

From the Theorem 40 it follows that the circle c will partition the circle q into two mutually inverse arcs; the inner arc, that is located completely inside of the circle c , and the outer arc, that is located completely outside of the circle c (Figure 9.3):



We already know that due to the Theorem 1, the points where the circles c and q intersect, P_1 and P_2 , will be fixed, since the power of inversion in question is positive.

Hence, under the inversion with respect to the circle c with positive power any point, such as the point A or B , that is located on the inner arc of the circle q will be transformed into a point, such as A' or B' ,

that is located on the outer arc of the circle q and vice versa; any point, such as the point A' or B' , that is located on the outer arc of the circle q will be transformed into a point, such as the point A or B , that is located on the inner arc of the circle q .

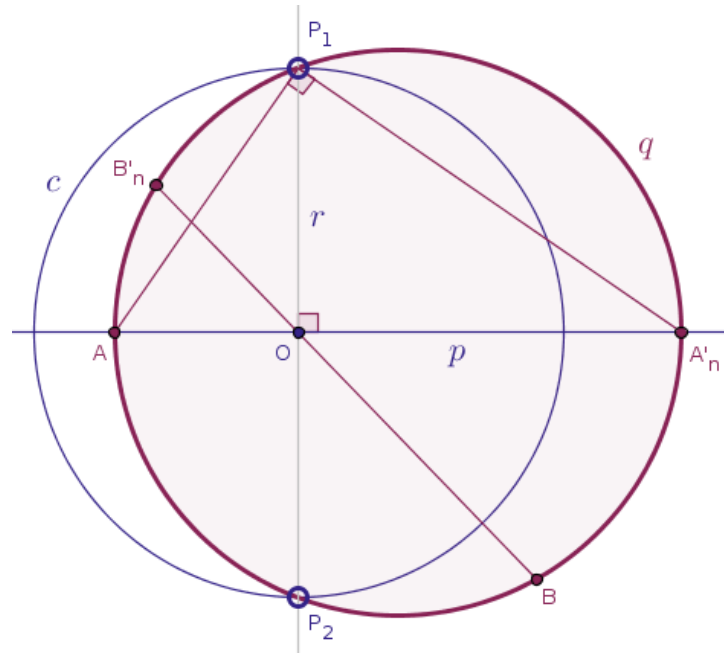
It follows, then, that even though such image points on the circle q do not transform into themselves, the circle q itself as a whole - does. This proves that the circle q is invariant under the inversion of the parent plane with positive power with respect to the circle c , which, by the way, is yet another example of **an invariant figure** and of **an identity transformation**.

By swapping the roles of the circles c and q and repeating the above couple of paragraphs almost verbatim, we prove that the circle c is invariant under the inversion of the parent plane with positive power with respect to the circle q . $\square$

Observe that the Theorems 39, 40 and 41 apply only to a corresponding inversion with **positive** power. In the next theorem the circles under investigation will, of course, not be orthogonal but for inversions with **negative** power that theorem, in some intuitive sense, is very similar in its spirit to that of the Theorem 41.

Theorem 42. Any circle, q , that passes through two diametrically opposite points, P_1 and P_2 , located on another circle, $c(O, r)$, is invariant under the inversion of the parent plane with respect to that other circle, c , with **negative** power.

Proof. Construct a straight line through the points P_1 and P_2 (Figure 9.4):



and a straight line, p , through the point O perpendicular to the straight line P_1P_2 . Name the points where the straight line p and the circle q intersect as A and A'_n .

Notice that the circumference angle $AP_1A'_n$ subtends the diameter AA'_n of the circle q and, hence, by [Euclid's "Elements" Book 3 Proposition 31](#), is right.

In the triangle $AP_1A'_n$ the straight line P_1P_2 is a perpendicular drawn from the right interior angle at the vertex P_1 to the base AA'_n . From [Euclid's "Elements" Book 6 Proposition 8](#) it follows, then, that the triangles AOP_1 and $P_1OA'_n$ are similar and, hence, by the nearby [Book 6 Proposition 4](#), it will be the case that:

$$\frac{|OA|}{|OP_1|} = \frac{|OP_1|}{|OA'_n|}$$

from where:

$$|OA| \cdot |OA'_n| = |OP|^2 = r^2$$

From the Definitions 1 and 3 it follows, then, that the points A and A'_n are the inverses of each other with respect to the circle c with **negative** power.

Moreover, pick an arbitrary point B on the circle q and construct a straight line through the points B and O until such a straight line intersects the circle q at the point B'_n . Applying [Euclid's "Elements" Book 3 Proposition 35](#) to the intersecting chords AA'_n and B'_nB , we find that:

$$|OB| \cdot |OB'_n| = |OA| \cdot |OA'_n| = r^2$$

which implies that the points B and B_n are the inverses of each other with respect to the circle c with negative power as well.

We, thus, again see the familiar pattern of the Theorem 41 when the circle c partitions the circle q into two mutually inverse arcs, an inner one and an outer one. Consequently, the circle q that passes through two diametrically opposite points, P_1 and P_2 , located on the circle $c(O, r)$ is invariant under the inversion of the parent plane with respect to the circle c with negative power.

Note carefully, however, that this time the circle c is not invariant under the inversion of the parent plane with respect to the circle q with negative power. $\square$

Theorem 43 (the converse of the Theorem 42). If a circle, q , is invariant under the inversion of a plane with respect to a circle $c(O, r)$ with negative power then the circle q intersects the circle c at two points, P_1 and P_2 , that are located diametrically opposite to each other on c .

Proof. Reusing the diagram of Figure 9.4 above, we state that since the circle q is invariant under the inversion of the parent plane with respect to the circle c with negative power, **any** (!) pair of points on it must trade places, while remaining collinear with the point O .

The points of intersection of the circles q and c , namely the points P_1 and P_2 , form such a pair and, since they too must be collinear with the point O , they must be located diametrically opposite to each other on the circle c . $\square$

Now that we know that the angles between and the tangency points of the curves under an inversion of the real plane with respect to a circle are preserved, we can investigate how the centers of the inverse circles are connected to each other in reference to the center of inversion. To that end, we solve the following problem.

Problem 10. Given a circle of inversion, $c(O, r)$, and two circles, $q(Q, R)$ and $q'(Q', R')$, which are the inverses of each other with respect to the circle c , find a way to invert the center of one of these circles, Q , into the center, Q' , of the other circle (and vice versa).

Solution. Let us consider an inversion of the parent plane with positive power first.

We solve this problem in reverse order by assuming temporarily that both centers, Q and Q' , are already and somehow available to us.

From the Definition 4 it follows that any straight line passing through the point Q will be orthogonal to the circle q .

Out of the multitude of such straight lines, let us pick a convenient one that does not pass through the center of inversion, O .

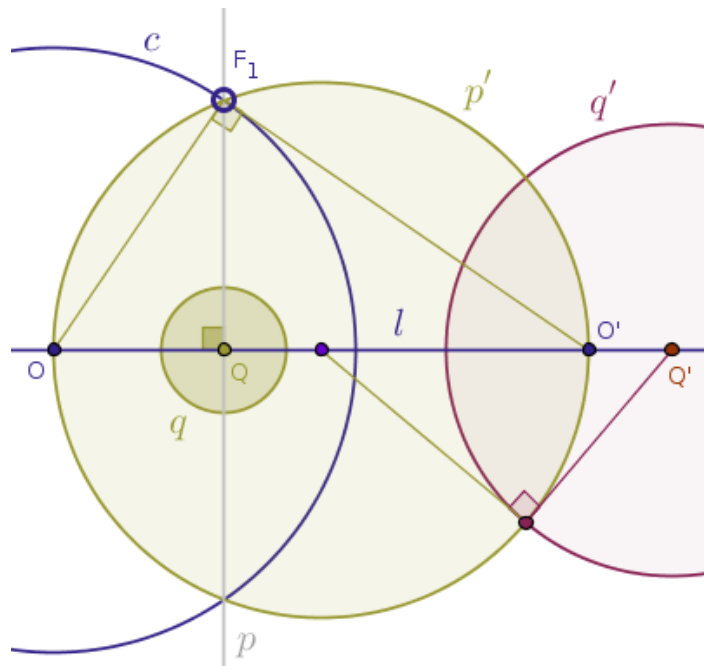
For example, let us pick a straight line, p , that is perpendicular to the straight line, l , passing through the points O , Q and Q' .

We already know that the magnitudes of angles between the original curves and their images under an inversion of the real plane are preserved.

Therefore, if the straight line p is orthogonal to the circle q then the image of p with respect to c with positive power, name it p' , will be orthogonal to the given image q' of q with respect to c with positive power.

However, in this case, from the Theorem 10 it follows that the image p' will be a circle that passes through the center of inversion, O , and intersects the straight line l at the point O' .

Since p' is orthogonal to q' , from the Theorem 41 it follows that p' will be invariant under the inversion of the parent plane with respect to the image circle q' with positive power, which, in turn, means that the point O' will be the image of the point O with respect to q' under that same inversion (Figure 9.5):



From our construction of the image circle p' , see the Theorem 10, it follows that the point O' will also be the image of the point Q with respect to the circle c (with positive power). As such, the point O' plays two roles at the same time:

- the point O' is the inverse of Q with respect to c with positive power and
- the point O' is the inverse of O with respect to q' with positive power also

Thus, the solution to our problem should be obvious now. In order to locate the center of the circle q , the point Q , given the center, the point Q' , of the circle q' that is the inversive image of the circle q with respect to the circle c with positive power:

- first, invert the point O with respect to the circle q' with positive power into the point O' and then
- invert the point O' with respect to the circle c also with positive power into the point Q

We can now reverse the order of the above operations and formulate the solution for locating the point Q' , given the location of the point Q in the forward flow of causality or in the sought-after order:

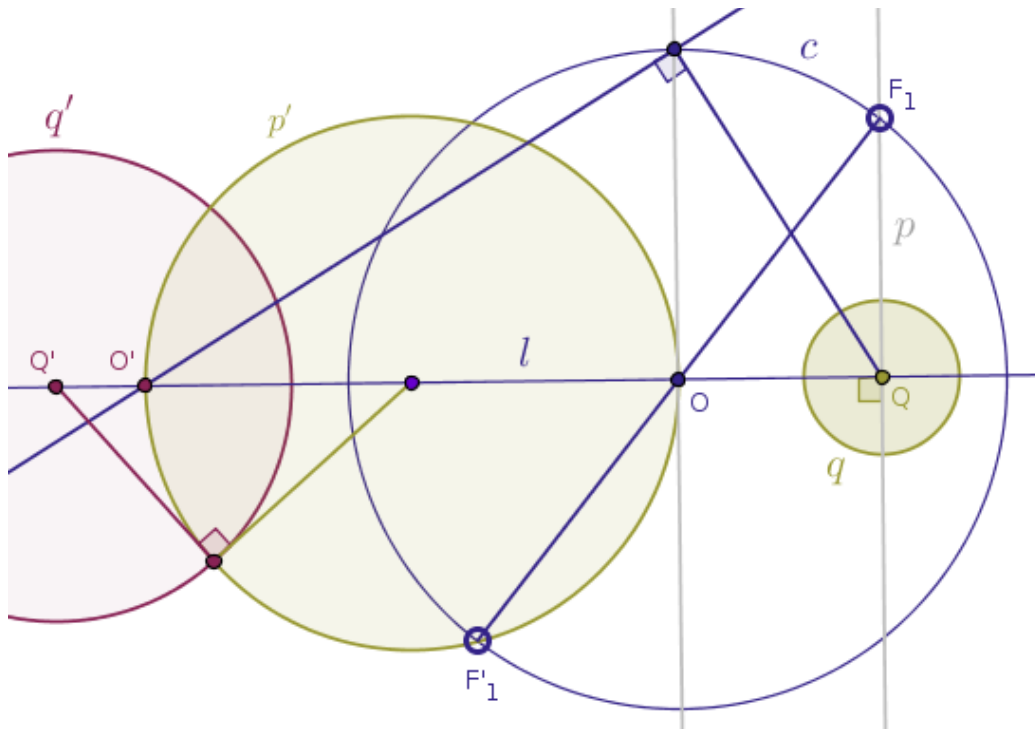
- first, invert the point O with respect to the circle q with positive power into the point O' and then
- invert the point O' with respect to the circle c also with positive power to locate the point Q'

Our readers are encouraged to generate the relevant construction for the latter case by themselves.

Now let us consider an inversion with negative power.

Here, likewise, a straight line, p , perpendicular to the straight line, l , passing through the points O , Q and Q' will always be orthogonal to the circle q . That straight line p will intersect the circle c at a point F_1 .

Hence, the image of the straight line p under the inversion of the parent plane with respect to the circle c with negative power, according to the Theorem 11, will be a circle p' that will pass through the center of inversion, O , and the point F'_1 diametrically opposite to the point F_1 . Moreover, the image circle p' will intersect the straight line l at the point O' that is diametrically opposite to the center of inversion (Figure 9.6):



But then the circle p' will be orthogonal to the image circle q' and, by the Theorem 41, the circle p' will be invariant under the inversion of the parent plane with respect to the image circle q' with **positive** (!) power, which, in turn, implies that the point O' will be the image of the point O under that same inversion. As such, we see that the point O' plays two roles at the same time:

- the point O' is the inverse of the point Q with respect to the circle c with **negative** power and
- the point O' is the inverse of the point O with respect to the circle q' with **positive** power

Thus, the solution to our problem should be obvious now. In order to locate the point Q , given the point Q' :

- first, invert the point O with respect to the circle q' with **positive** (!) power into the point O' and then
- invert the point O' with respect to the circle c with **negative** (!) power into the point Q

We can now reverse the order of the above operations and formulate the solution for locating the point Q' , given the point Q in the forward flow of causality or in the sought-after order:

- first, invert the point O with respect to the circle q with **positive** (!) power into the point O' and then
- invert the point O' with respect to the circle c with **negative** (!) power to locate the point Q'

Our readers are encouraged to generate the relevant construction for the latter case by themselves.

The Power Of A Point

The power of a point with respect to a circle defines a measure of the relative distance of a given point to a given, coplanar, circle as follows.

Definition 6. The power $\mathcal{P}$ of a point, A , with respect to a circle, $q(Q, r)$, is the difference between the square of the distance, d , from that point, A , to the center of that circle, Q , and the square of the radius of that circle, r :

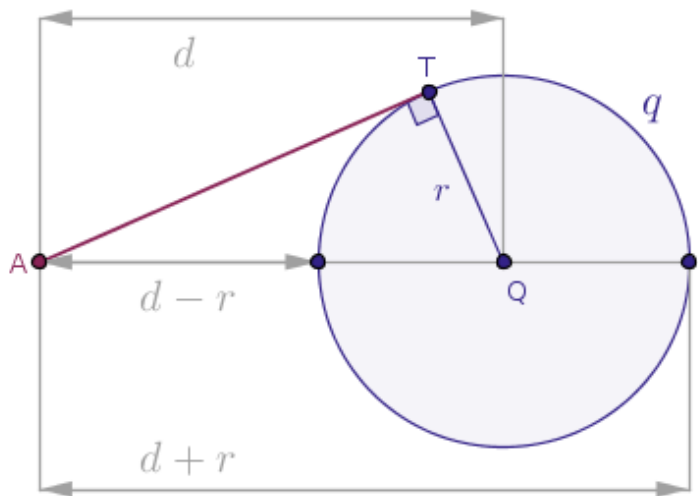
$$\mathcal{P}(A, q) = d^2 - r^2 = |AQ|^2 - r^2$$

We, thus, consider three distinct cases when the point of interest, A , is located outside of, on or inside of the given circle, q .

Case 1. The given point, A , is **outside** of the given circle, q .

If the given point, A , is located outside of the given circle, q , then it is always possible to draw a tangent straight line, AT , to that circle, q , through the given point, A , as shown in Figure 9.7 on the right.

Moreover, in this case, clearly, the distance from the given point to the center of the given circle is strictly larger than the radius



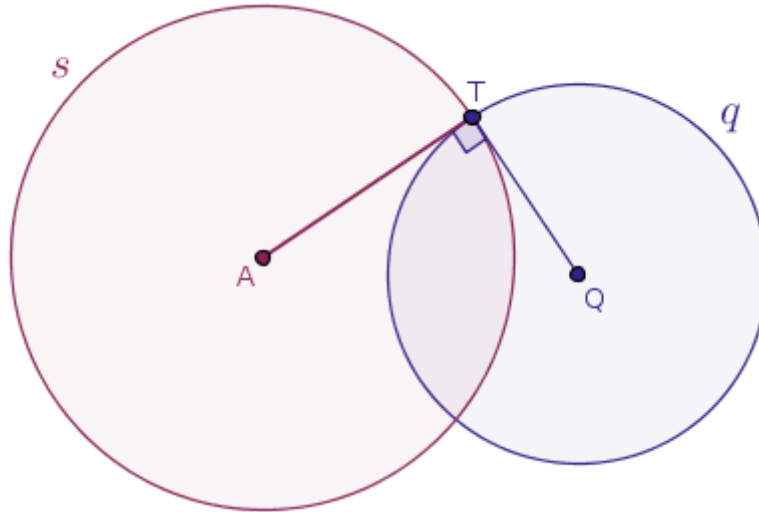
of that circle, $|AQ| = d > r$, which implies that the difference of the squares in the Definition 6 will always be a strictly positive real number and that difference can be rearranged it as follows:

$$\mathcal{P}(A, q) = d^2 - r^2 = (d - r)(d + r) = |AQ|^2 - r^2$$

The sum $(d + r)$ above is the length of the entire secant of the given circle, q , through the given point, A , while the difference $(d - r)$ above is the length of its outer portion only.

Hence, by the familiar to us Euclid's "Elements" Book 3 Proposition 36, the above formula follows right away and we observe that in this particular case the power of the given point, A , with respect to the given circle, q , is the square of the length of the tangent, AT , drawn to the given circle, q , through the given point, A .

Further, if we construct a circle, s , centered at the given point, A , with the radius whose length is equal exactly to the length of the said tangent, AT (Figure 9.8):



then the radius AT of the circle s will, by definition, be perpendicular to the radius QT of the circle q , implying that the circles s and q will be **orthogonal**. We will use that observation later on in the **Radical Axis** section below.

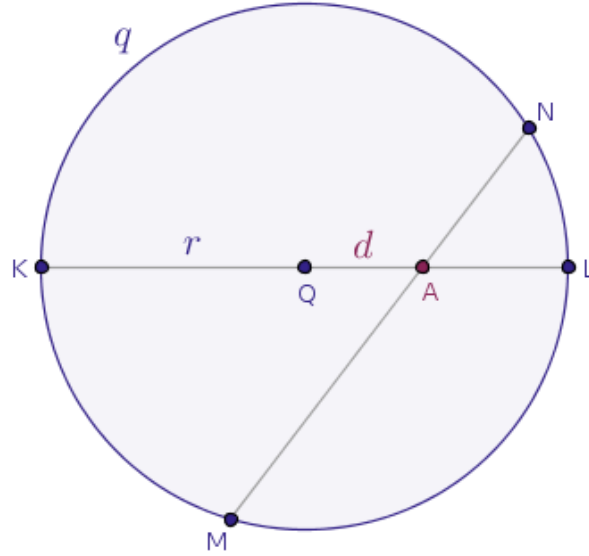
Case 2. The given point, A , is on the given circle, q .

If the given point A is located on the given circle, q , then, clearly, it is the case that the distance from that point to the center of the given circle, d , is equal to the radius of that circle, r , as in $d = r$, and, vanishing, the power of the given point, A , with respect to the given circle, q , in that case is equal to zero:

$$\mathcal{P}(A, q) = r^2 - r^2 = 0$$

Case 3. The given point, A , is inside of the given circle, q .

If the given point, A , is located inside of the given circle, q , then it is always possible to draw two secants of the given circle, q , say, the secants MN and KL , that intersect at the given point, A (Figure 9.9):



Moreover, it is evident that in this case the distance from the given point to the center of the given circle will be strictly less than the radius of that circle: $|AQ| = d < r$, which implies that the difference of the squares in the Definition 6 will always be a strictly negative real number and that difference can be rearranged it as follows:

$$\mathcal{P}(A, q) = d^2 - r^2 = -(r^2 - d^2) = -(r + d)(r - d) = -|AK| \cdot |AL| = -|AM| \cdot |AN|$$

which holds true for any secant MN of the circle q due to the Euclid's "Elements" Book 3 Proposition 35 also known as the intersecting chords theorem.

Radical Axis

We now have all the ducks in a row in order to introduce one more useful notion.

Definition 7. The radical axis of two coplanar circles, q_1 and q_2 , is a locus of points whose powers with respect to both of these circles, q_1 and q_2 , are equal.

Observe that strictly speaking there is no need for any symbols in the above definition. That is, it is entirely possible to define the radical axis of two coplanar circles without any symbols as follows. The radical axis of two coplanar circles is a set of points the power of each of which with respect to these two

circles is one and the same. However, the reason why we use some symbols in the above definition of the radical axis is to set up a context for the next theorem and to help our readers digest this definition a bit further on their own.

In more detail, what the Definition 7 says is the following. Say, we are given two coplanar circles, a circle named q_1 and a distinct and coplanar circle named q_2 . If we take any point A in the parent plane and compute the power of that point with respect to the circle q_1 then it is highly unlikely that that real number will match the power of that same point A with respect to the second circle, the circle q_2 .

However, from the very definition of the notion of radical axis of two coplanar circles it is evident that it is possible to locate some number of points in the parent plane such that if the powers of any one such point is computed with respect to both circles q_1 and q_2 then these powers will match:

$$\mathcal{P}(A, q_1) = \mathcal{P}(A, q_2)$$

The set of points in the parent plane for all of which the above equality holds if continued across all of these points:

$$\mathcal{P}(A_2, q_1) = \mathcal{P}(A_2, q_2) \quad \text{AND} \quad \mathcal{P}(A_3, q_1) = \mathcal{P}(A_3, q_2) \quad \text{AND} \quad \dots$$

is called exactly the radical axis of these two circles and the next theorem investigates all the reasonable cases when the radical axis of two coplanar circles exists. In the section that follows the proof of that theorem we look at practical ways of constructing radical axes with Euclidean tools alone in each case.

Theorem 44. The radical axis of any two non-concentric coplanar circles, $q_1(Q_1, r_1)$ and $q_2(Q_2, r_2)$, is a straight line perpendicular to the straight line passing through the centers of these circles, the points Q_1 and Q_2 .

Proof. We can always choose an orthogonal xOy coordinate system in such a way the its x -axis passes through the centers of the given circles, the points Q_1 and Q_2 , and its y -axis passes through the middle point, O , that is located in between these centers, Q_1 and Q_2 , meaning that $|Q_1O| = |Q_2O| = a$.

Next, in the xOy plane we can place an arbitrary point, A , anywhere with respect to the coordinate axes and the centers Q_1 and Q_2 , but regardless of this placement, the proof of this theorem follows the same, admittedly algebraic, path.

Let d_1 and d_2 be the distances from such an arbitrary point A to the centers Q_1 and Q_2 correspondingly. By definition of a power of a point with respect to a circle, in this case we have:

$$\mathcal{P}_1 = \mathcal{P}(A, q_1) = d_1^2 - r_1^2$$

$$\mathcal{P}_2 = \mathcal{P}(A, q_2) = d_2^2 - r_2^2$$

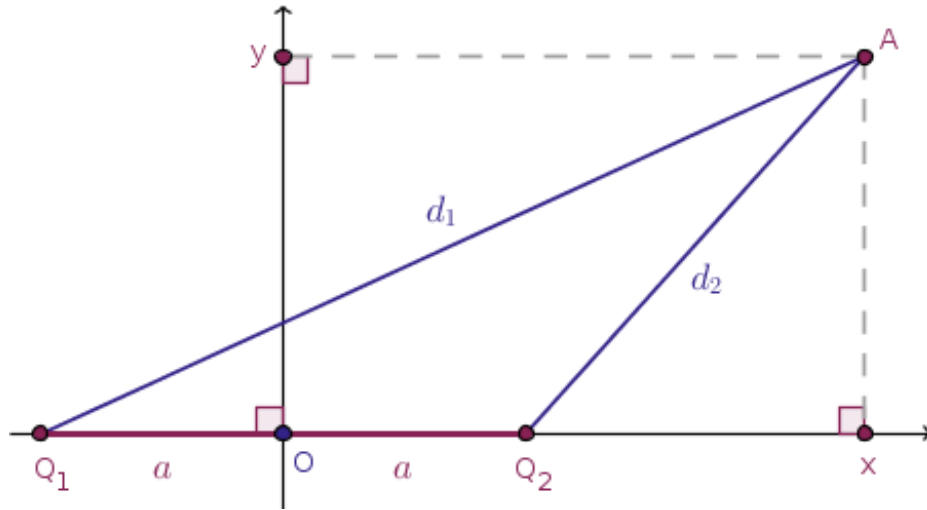
By the definition of the radical axis given earlier, we have $\mathcal{P}_1 = \mathcal{P}_2$. Hence:

$$d_1^2 - r_1^2 = d_2^2 - r_2^2$$

and, since all the numbers $d_{1,2}$ and $r_{1,2}$ remain constant at all times, it follows that the following magnitude is itself constant:

$$d_1^2 - d_2^2 = r_1^2 - r_2^2 = k = \text{const}$$

We may further always assume that $k \geq 0$, since otherwise we simply swap the roles of the circles q_1 and q_2 and renumber them (Figure 9.10):



Let the variable coordinates of an arbitrary point A in the plane xOy be (x, y) . Then, from the right, by construction, triangles AxQ_1 and AxQ_2 we have:

$$d_1^2 = y^2 + (a + x)^2 = y^2 + x^2 + 2ax + a^2$$

$$d_2^2 = y^2 + (x - a)^2 = y^2 + x^2 - 2ax + a^2$$

which means that:

$$d_1^2 - d_2^2 = 4ax = k$$

from where:

$$x = \frac{k}{4a}$$

Again, since all the magnitudes on the RHS of the above equation remain constant at all times, it follows that that equation represents the equation of a straight line perpendicular to the x -axis. Therefore, the radical axis of any two non-concentric circles, $q_1(Q_1, r_1)$ and $q_2(Q_2, r_2)$, is a straight line perpendicular to the straight line passing through the centers of these circles, the points Q_1 and Q_2 . $\square$

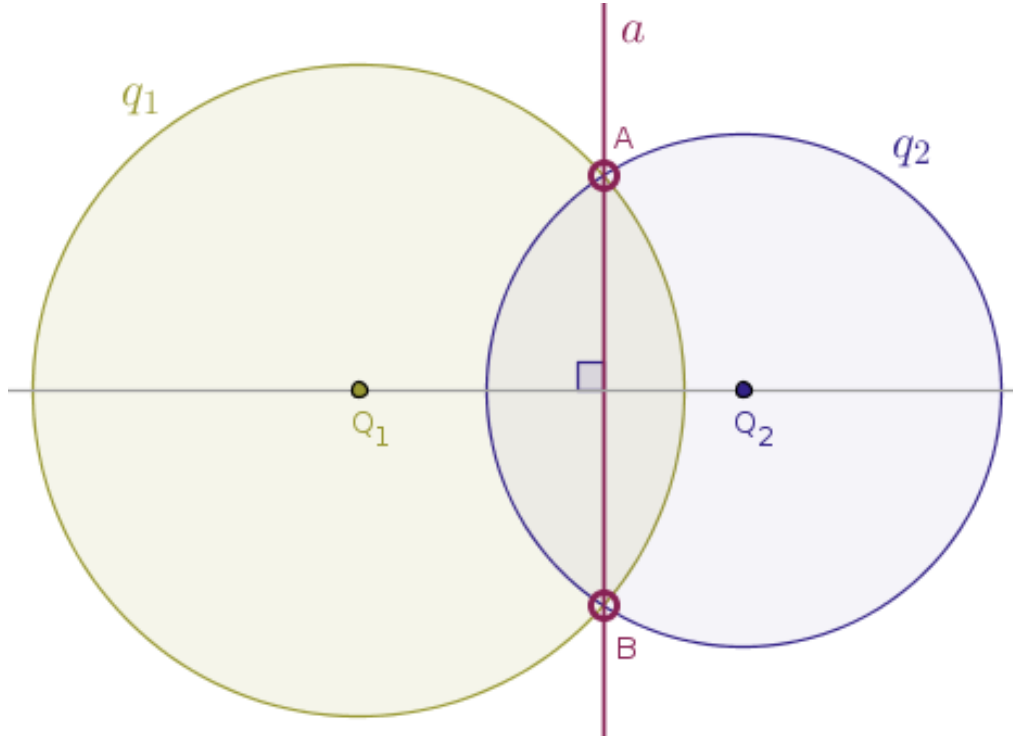
Radical Axis Construction

On the practical side, note that from the above theorem it follows that the construction of the radical axis of two non-concentric coplanar circles is reduced to the recovery of the location of a point equivalent to the magnitude $k/4a$ shown above.

As such, we consider three cases for the construction of the radical axis, a , of two non-concentric circles that belong to one and the same plane.

Case 1. Two Non-concentric Coplanar Circles Intersect At Two Distinct Points.

Let any two non-concentric co-planar circles, $q_1(Q_1, r_1)$ and $q_2(Q_2, r_2)$, intersect at two distinct points, A and B (Figure 9.11):



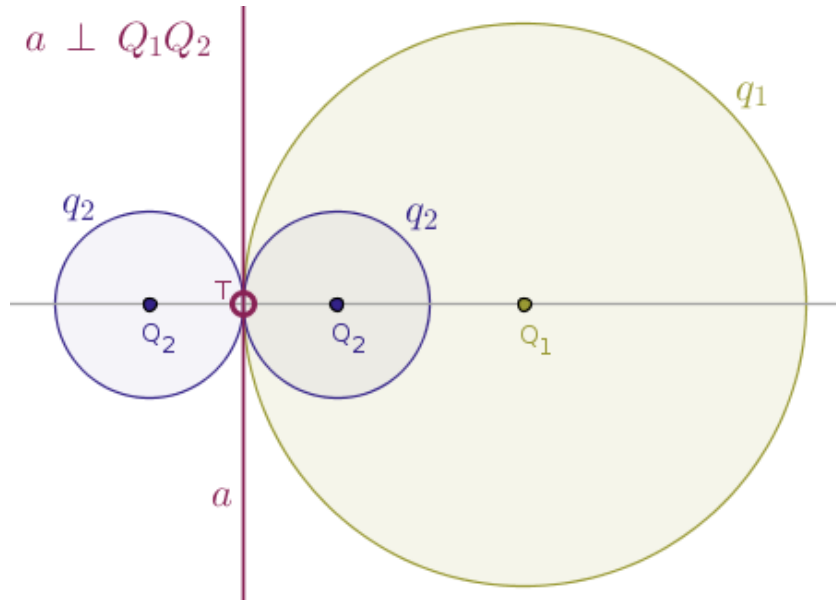
From the definition of the radical axis and the Theorem 44 it follows that in that case the sought-after radical axis, a , of the intersecting circles q_1 and q_2 , must be a straight line that passes precisely through the points, A and B , that these two circles have in common because:

- the powers of both of these points with respect to the given circles are equal to zero and, hence,
- are equal to one another and
- two distinct points define a unique straight line, which is also perpendicular to the straight line of centers of the given circles

Our readers should either recall or prove on their own that in any two intersecting coplanar circles the chord drawn through the two points that are common to these two circles is perpendicular to the line of centers of these circles.

Case 2. Two Non-concentric Coplanar Circles Touch.

Let the two non-concentric coplanar circles, $q_1(Q_1, r_1)$ and $q_2(Q_2, r_2)$, touch internally or externally at a single point, T (Figure 9.12):



From the definition of the radical axis and the Theorem T44 it follows that in that case the sought-after radical axis, a , of the tangent circles q_1 and q_2 must be a tangent straight line drawn to both circles through the point of tangency, T , because:

- the powers of the tangency point, T , with respect to the given circles are equal to zero and, hence,
- are equal to one another and

- a tangent straight line in that case is unique and it is also perpendicular to the straight line of the centers, Q_1 and Q_2

Again, here our readers should either recall or prove on their own the fact that through a point on a straight line there passes one and only one straight line that is perpendicular to it.

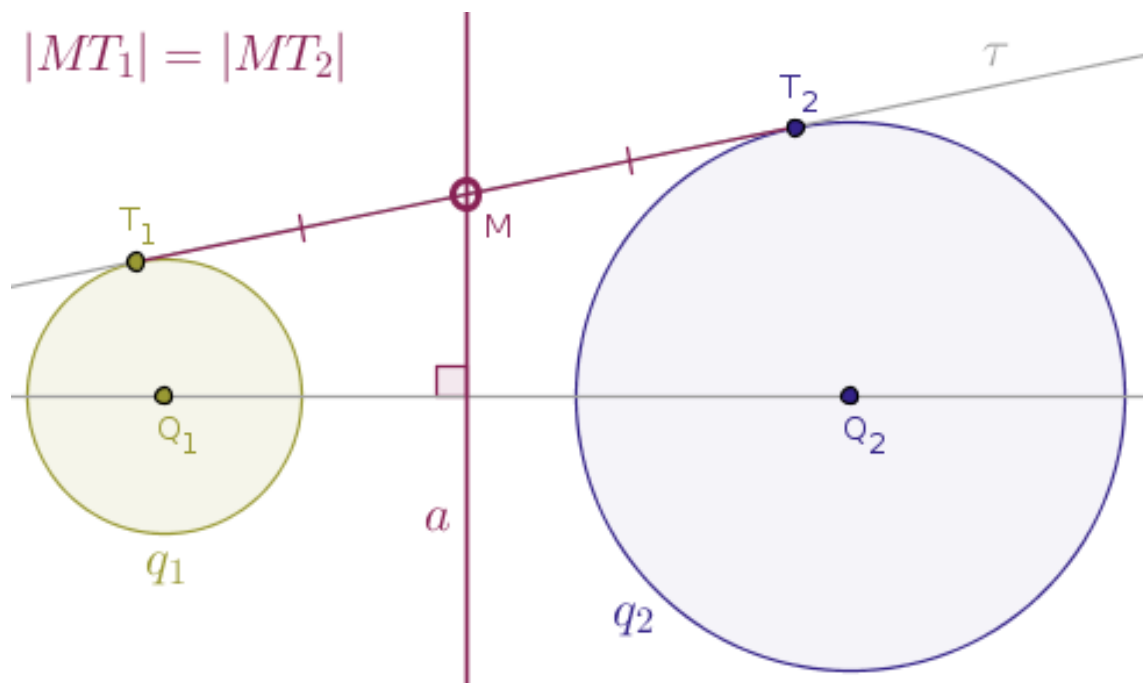
Case 3. Two Non-concentric Coplanar Circles Have Zero Points In Common.

First, let any two such circles, $q_1(Q_1, r_1)$ and $q_2(Q_2, r_2)$, that have exactly zero common points be located **outside** of each other.

Then, there is more than one way to recover the radical axis of two such circles and the easier one would be based on the observation that the power of a point with respect to a circle is the square of the length of a tangent straight line drawn to the circle through the said point.

Thus, in order to find the radical axis, a , of two circles that have no common points and are located outside of each other, we need to find a mystery point M the length of a tangent from which to both circles is one and the same.

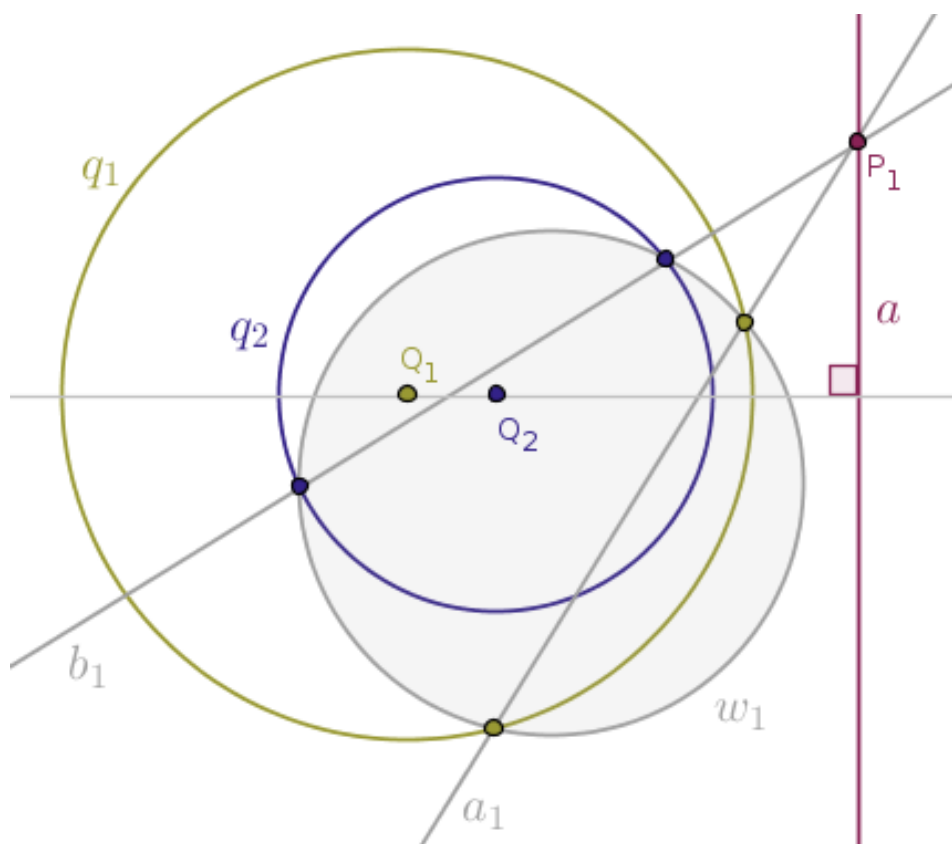
One such obvious point, for example, is the middle point M of a common to both circles **external** tangent straight line, τ . Constructing a straight line through such a point M perpendicular to the straight line of the centers Q_1 and Q_2 locates the radical axis a sought-after (Figure 9.13):



Next, let the two non-concentric coplanar circles, $q_1(Q_1, r_1)$ and $q_2(Q_2, r_2)$, that have exactly zero common points be located one **inside** of the other. Without the loss of generality let q_2 be located inside of q_1 .

In this case it is easier to construct the radical axis, a , of the circles q_1 and q_2 by employing the method of intersecting loci and by capitalizing on the fact that we already know that the radical axis of two circles that do intersect, over two distinct points, passes through the said points of intersection, see the Case 1 above.

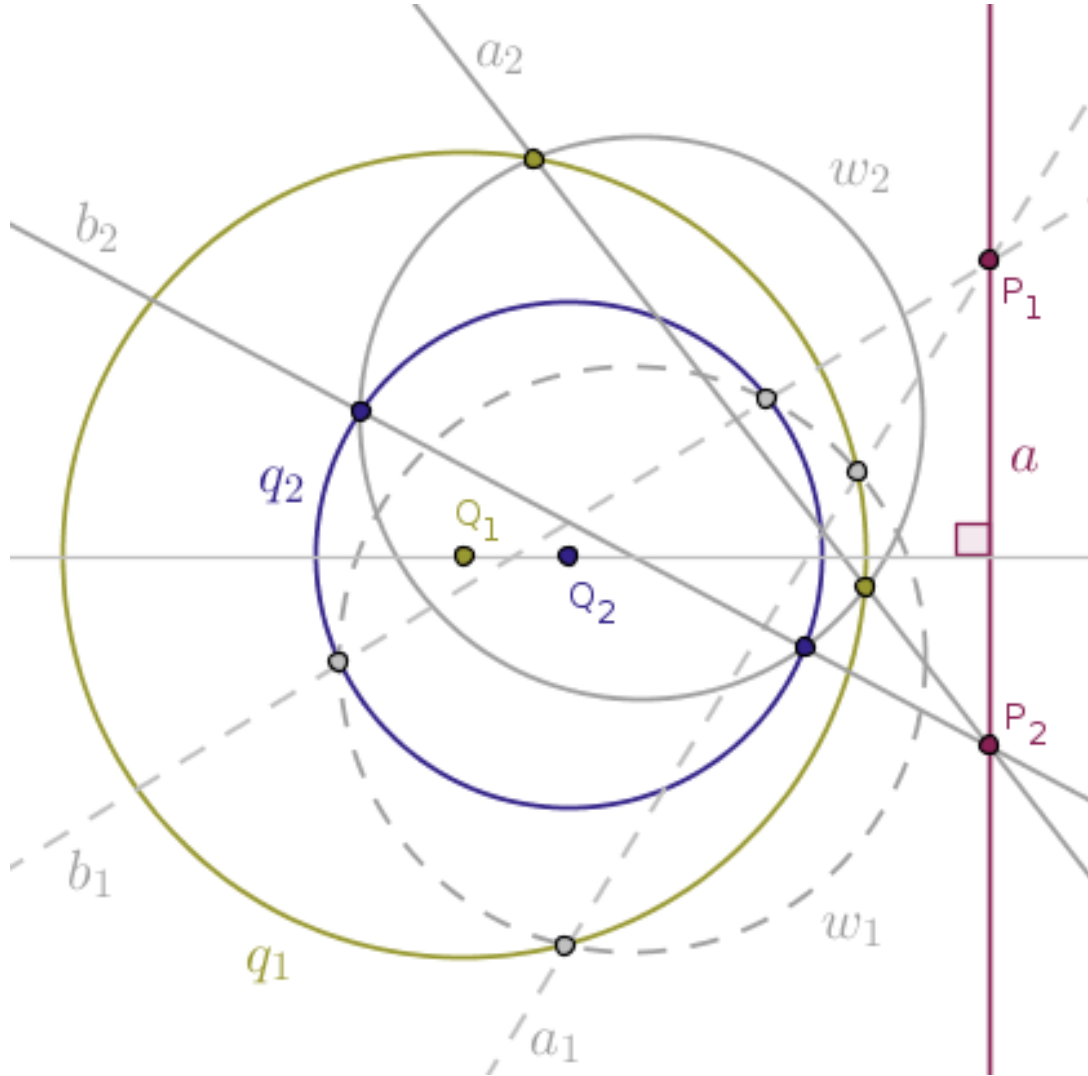
As such, if we draw an arbitrary circle, w_1 , that intersects both circles q_1 and q_2 at two distinct points each then we will obtain the radical axis, a_1 , of the circles w_1 and q_1 and the radical axis b_1 of the circles w_1 and q_2 . It stands to reason, then, that the point, P_1 , where the radical axes a_1 and b_1 intersect must also belong to the sought-after radical axis, a , of the circles q_1 and q_2 (Figure 9.14):



Now there are two options.

On the one hand, it is possible to directly construct a straight line, a , perpendicular to the straight line of centers, Q_1Q_2 , passing through the point P_1 as is shown in the diagram above.

On the other hand, it is possible to draw a distinct from w_1 circle w_2 that also intersects both circles q_1 and q_2 at two distinct points each and, thus, obtain the radical axis, a_2 , of the circles w_2 and q_1 and the radical axis, b_2 , of the circles w_2 and q_2 . These two radical axes, a_2 and b_2 , will intersect at the point $P_2 \neq P_1$ that will also belong to the radical axis, a , of the circles q_1 and q_2 (Figure 9.15):



Hence, a unique straight line, a , through the distinct points P_1 and P_2 will be the radical axis, a , sought-after. $\square$

As the conclusion of this topic, let us consider certain points located exactly **on** the radical axis.

Some of these points can be located **inside** of both circles. In the Case 1 above, for example, such points will be in between the intersection points A and B .

Some of these points can be located **on** both circles. In the case Case 2 above, for example, such a point could be the point of tangency of two circles, T .

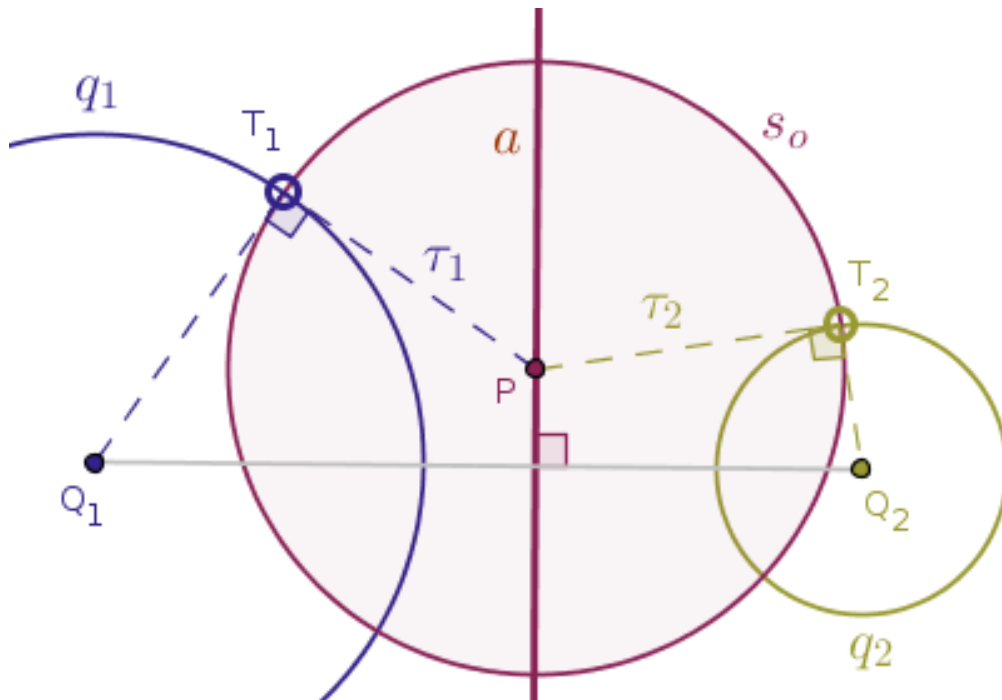
Some of these points can be located **outside** of both circles, as in all the other cases.

When a certain point on a radical axis is located outside of (or on) both circles then we can apply the Power of a Point, Case 1 observation made earlier: any circle centered at such a point with the radius equal to the length of the tangent straight line drawn to either circle through that point will be orthogonal to both given circles simultaneously.

When a point on a radical axis is located on both given circles, as the point T in the Case 2 above, then we simply have a case of a degenerate circle of zero radius.

It follows, then, that for any point on a radical axis located on or outside of both given circles there exists **a unique circle orthogonal to both** of them. In other words, the locus of the centers of circles orthogonal to both given circles is the entire radical axis for the Cases 2, 3 above and the entire radical axis excluding the line segment AB for the Case 1.

In order to construct one such circle, s_o , choose a point, P , on the radical axis that is located outside of both given circles, construct a tangent straight line, τ_1 (or τ_2) through the point P to the circle q_1 (or q_2) in order to locate their tangency point, T_1 (or T_2), draw a circle, s_o , centered at the point P with the radius PT_1 (or PT_2) (Figure 9.16):



and so on.